

ECON 5345 Homework 9 Report

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1. The endogenous variables are

$$x_t = \begin{bmatrix} \hat{c}_t & \hat{i}_t & \hat{y}_t & \hat{l}_t \end{bmatrix}'.$$

The state variables are

$$s_t = \begin{bmatrix} \hat{k}_{t-1} & \hat{z}_t \end{bmatrix}'.$$

The linearized equilibrium conditions are

$$\sigma(\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t) = \beta R^* \mathbb{E}_t(\hat{y}_{t+1} - \hat{k}_t) + \beta \phi \mathbb{E}_t(\hat{k}_{t+1} - \hat{k}_t) - \phi(\hat{k}_t - \hat{k}_{t-1}), \quad (1)$$

$$\left(1 + \frac{1}{\eta}\right) \hat{l}_t = -\sigma \hat{c}_t + \hat{y}_t, \quad (2)$$

$$\hat{y}_t = \hat{z}_t + \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{l}_t, \quad (3)$$

$$\hat{y}_t = \frac{C^*}{Y^*} \hat{c}_t + \frac{I^*}{Y^*} \hat{i}_t, \quad (4)$$

$$\hat{k}_t = (1 - \delta) \hat{k}_{t-1} + \delta \hat{i}_t, \quad (5)$$

$$\hat{z}_t = \rho \hat{z}_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2). \quad (6)$$

For a given $\theta = (\rho, \phi, \sigma_\varepsilon)$, solving the linear rational expectations model yields the reduced-form decision rule

$$x_t = H(\theta) s_t,$$

where

$$H(\theta) = \begin{bmatrix} h_{ck}(\theta) & h_{cz}(\theta) \\ h_{ik}(\theta) & h_{iz}(\theta) \\ h_{yk}(\theta) & h_{yz}(\theta) \\ h_{lk}(\theta) & h_{lz}(\theta) \end{bmatrix}.$$

Then the capital accumulation equation becomes

$$\hat{k}_t = (1 - \delta)\hat{k}_{t-1} + \delta\hat{i}_t \quad (7)$$

$$= \left[(1 - \delta) + \delta h_{ik}(\theta) \right] \hat{k}_{t-1} + \delta h_{iz}(\theta) \hat{z}_t. \quad (8)$$

Thus the state transition equation is

$$s_{t+1} = \begin{bmatrix} \hat{k}_t \\ \hat{z}_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} (1 - \delta) + \delta h_{ik}(\theta) & \delta h_{iz}(\theta) \\ 0 & \rho \end{bmatrix}}_{B(\theta)} s_t + \underbrace{\begin{bmatrix} 0 \\ \sigma_\varepsilon \end{bmatrix}}_{u_{t+1}} w_{t+1}. \quad (9)$$

where $w_{t+1} \sim N(0, 1)$ is the system error.

The observed data are measured with error:

$$x_t^* = x_t + e_t, \quad e_t \sim N(0, \Xi), \quad \Xi = \text{diag}(0.25, 0.25, 0.25, 0.25).$$

Therefore the measurement equation is

$$x_t^* = H(\theta)s_t + e_t. \quad (10)$$

Hence the state-space representation is

$$s_{t+1} = B(\theta)s_t + u_{t+1}, \quad (11)$$

$$x_t^* = H(\theta)s_t + e_t. \quad (12)$$

We do not need to recompute the steady state for each candidate θ because ρ and σ_ε affect only the shock process, while ϕ affects only transition dynamics. They are all irrelevant to the steady state.

2. We first solve for the log-linearized system.

Write $b_{kk}(\theta) = (1 - \delta) + \delta h_{ik}(\theta)$ and $b_{kz}(\theta) = \delta h_{iz}(\theta)$. In the following, we drop the dependence on θ for notational clarity.

We then have

$$\hat{k}_t = b_{kk}\hat{k}_{t-1} + b_{kz}\hat{z}_t, \quad (13)$$

$$\hat{c}_t = h_{ck}\hat{k}_{t-1} + h_{cz}\hat{z}_t \quad (14)$$

Combining (2) and (3), we get

$$\hat{\ell}_t = \frac{-\sigma c_t + z_t + \alpha k_{t-1}}{\alpha + 1/\eta}. \quad (15)$$

Substituting this into (3), we get

$$\hat{y}_t = \left(\frac{1 + 1/\eta}{\alpha + 1/\eta} \right) \hat{z}_t + \left(\frac{\alpha(1 + 1/\eta)}{\alpha + 1/\eta} \right) \hat{k}_{t-1} + \left(\frac{-\sigma(1 - \alpha)}{\alpha + 1/\eta} \right) \hat{c}_t.$$

By (13) and (14), this can be further reduced to

$$\hat{y}_t = \frac{\alpha(1 + 1/\eta) - \sigma h_{ck}(1 - \alpha)}{\alpha + 1/\eta} \hat{k}_{t-1} + \frac{(1 + 1/\eta) - \sigma h_{cz}(1 - \alpha)}{\alpha + 1/\eta} \hat{z}_t. \quad (16)$$

Combining (4) and (5), we get

$$\hat{y}_t = \frac{C^*}{Y^*} \hat{c}_t + \frac{K^*}{Y^*} [\hat{k}_t - (1 - \delta)\hat{k}_{t-1}].$$

By (13) and (14), this can be further reduced to

$$\hat{y}_t = \left[\frac{C^*}{Y^*} h_{ck} + \frac{K^*}{Y^*} [b_{kk} - (1 - \delta)] \right] \hat{k}_{t-1} + \left[\frac{C^*}{Y^*} h_{cz} + \frac{K^*}{Y^*} b_{kz} \right] \hat{z}_t. \quad (17)$$

Equating the coefficients of $\hat{k}_{t-1}$ and $\hat{z}_t$ in (16) and (17), we get

$$\frac{\alpha(1 + 1/\eta) - \sigma h_{ck}(1 - \alpha)}{\alpha + 1/\eta} = \frac{C^*}{Y^*} h_{ck} + \frac{K^*}{Y^*} [b_{kk} - (1 - \delta)], \quad (18)$$

$$\frac{(1 + 1/\eta) - \sigma h_{cz}(1 - \alpha)}{\alpha + 1/\eta} = \frac{C^*}{Y^*} h_{cz} + \frac{K^*}{Y^*} b_{kz}. \quad (19)$$

Two more equations can be obtained by the Euler equation (1). (13) and (14) imply

the following:

$$\begin{aligned}
\mathbb{E}_t \hat{c}_{t+1} &= h_{ck} k_t + \rho h_{cz} z_t, \\
\mathbb{E}_t \hat{y}_{t+1} &= \underbrace{\frac{\alpha(1 + 1/\eta) - \sigma h_{ck}(1 - \alpha)}{\alpha + 1/\eta}}_{y_k} k_t + \underbrace{\frac{(1 + 1/\eta) - \sigma h_{cz}(1 - \alpha)}{\alpha + 1/\eta}}_{y_z} \rho \hat{z}_t, \\
\mathbb{E}_t \hat{k}_{t+1} &= b_{kk} k_t + b_{kz} \rho z_t.
\end{aligned}$$

Substituting these together with (13) and (14) into (1), we get

$$\begin{aligned}
&\sigma \left[h_{ck}(b_{kk} - 1) \hat{k}_{t-1} + [h_{ck} b_{kz} + h_{cz}(\rho - 1)] \hat{z}_t \right] \\
&= \beta R^* \left[(y_k - 1) b_{kk} \hat{k}_{t-1} + [(y_k - 1) b_{kz} + y_z \rho] \hat{z}_t \right] \\
&\quad + \beta \phi \left[b_{kk}(b_{kk} - 1) \hat{k}_{t-1} + [b_{kk} b_{kz} + b_{kz}(\rho - 1)] \hat{z}_t \right] \\
&\quad - \phi \left[(b_{kk} - 1) \hat{k}_{t-1} + b_{kz} \hat{z}_t \right].
\end{aligned}$$

Equating the coefficients on both sides, we get two more equations:

$$\begin{aligned}
\sigma h_{ck}(b_{kk} - 1) &= \beta R^*(y_k - 1) b_{kk} + \beta \phi b_{kk}(b_{kk} - 1) - \phi(b_{kk} - 1), \quad (20) \\
\sigma [h_{ck} b_{kz} + h_{cz}(\rho - 1)] &= \beta R^*[(y_k - 1) b_{kz} + y_z \rho] + \beta \phi [b_{kk} b_{kz} + b_{kz}(\rho - 1)] - \phi b_{kz}. \quad (21)
\end{aligned}$$

Substituting the parameter values, we can get the solution for h_{ck} , h_{cz} , b_{kk} , and b_{kz} . As a result, we get h_{yk} and h_{yz} from (16), h_{lk} and h_{lz} from (15), and h_{ik} and h_{iz} from (5).

The simulated data are generated by the code. We simulated 2000 times and burnt the first 1000. We start the initial state with $s_0 = [0 \ 0]'$, that is, the steady state. Simulated data are saved in the file `hw9_partb_simulated_data.txt`. Here we report the H matrix and the B matrix.

$$H = \begin{bmatrix} 0.4738 & 0.5911 \\ -0.4139 & 3.1746 \\ 0.2631 & 1.2044 \\ -0.1054 & 0.3067 \end{bmatrix}, \quad B = \begin{bmatrix} 0.9647 & 0.0794 \\ 0 & 0.9 \end{bmatrix}.$$

3. I report the parameter estimates and covariance matrix.

Parameter	True Value	Estimate	Std. Error
ρ	0.9	0.9082	0.0007
ϕ	5.0	4.9420	0.0245
σ_ε	1.0	1.0134	0.0235

Table 1: Parameter Estimates

	ρ	ϕ	σ_ε
ρ	4.6×10^{-7}	-2.7×10^{-7}	4.7×10^{-7}
ϕ	-2.7×10^{-7}	6.0126×10^{-4}	3.373×10^{-5}
σ_ε	4.7×10^{-7}	3.373×10^{-5}	5.5353×10^{-4}

Table 2: Covariance Matrix of θ

The estimated state-space matrices are

$$\hat{H} = \begin{bmatrix} 0.4745 & 0.5943 \\ -0.4174 & 3.1575 \\ 0.2627 & 1.2028 \\ -0.1059 & 0.3043 \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} 0.9646 & 0.0789 \\ 0 & 0.9082 \end{bmatrix}.$$